



Arduino Water Turbidity Monitor

Sense • Measure • Display

INTRODUCTION

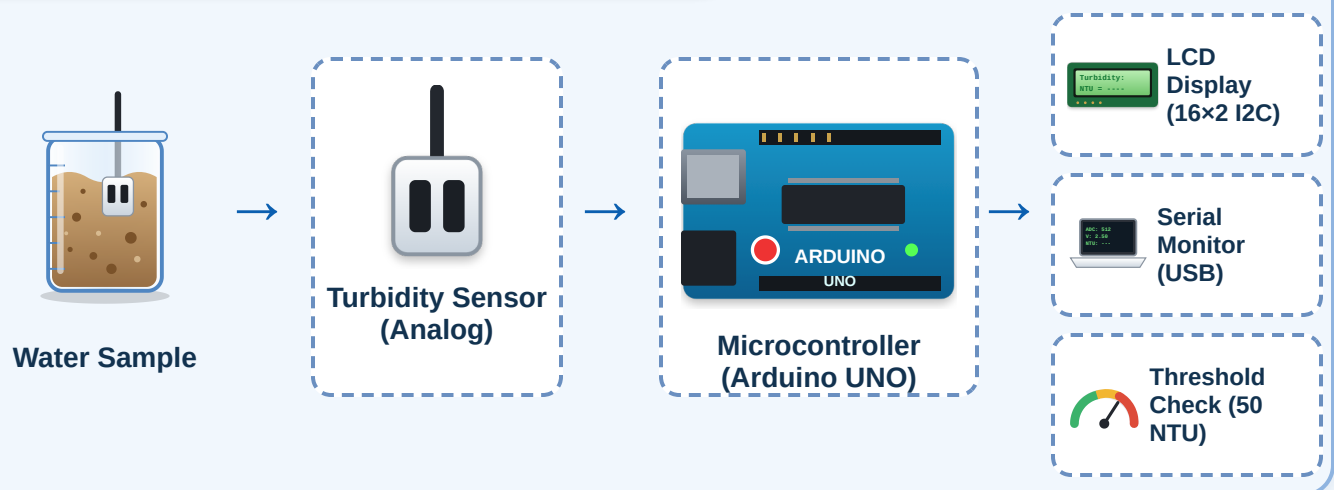
AQUA GUARD is a smart water turbidity monitoring system designed to measure how cloudy water is and display the result instantly. It reads an analog turbidity sensor, estimates NTU and shows the value on an LCD.



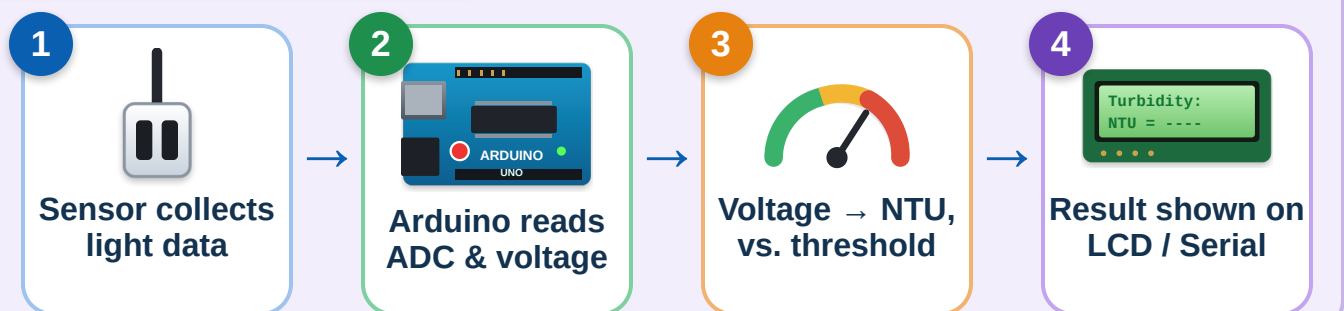
KEY FEATURES

- Turbidity Sensing**
Detects cloudiness of water using an analog sensor.
- NTU Estimation**
Converts sensor voltage to an estimated NTU value.
- Instant Display**
Shows readings on a 16×2 LCD and Serial Monitor.
- Configurable Threshold**
50 NTU limit set as a constant in code.
- Low Cost & Simple**
Compact build from easily available parts.

SYSTEM ARCHITECTURE



HOW IT WORKS



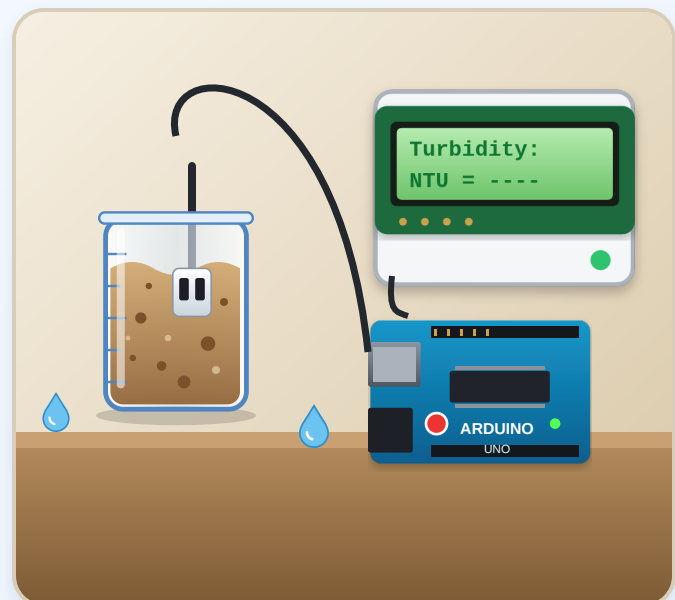
APPLICATIONS



BENEFITS

- ✓ Instant local water-clarity readout
- ✓ Low-cost and easy to build
- ✓ Threshold adjustable in code
- ✓ No lab equipment needed
- ✓ Ready for future IoT upgrade

Smart Sensing for Clearer Water



Prototype: NTU calibration & safe/unsafe classification are under validation; not a certified drinking-water verdict.



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Clean
Water
First